

END-TO-END fMRI PREPROCESSING MANUAL

From Siemens DICOM to BIDS and fMRIPrep 25.2.5 outputs

Student laboratory workflow for a shared Linux server



Version 1.0 | 28 July 2026

All user-specific storage paths are written under /your_folder/.

How to Use This Manual

This manual describes a reproducible, student-ready workflow for converting scanner DICOM files to BIDS, validating the dataset, and running fMRIPrep in Docker. It assumes that Docker Engine and the fMRIPrep image are already installed on a shared Linux server. Students create a Conda environment only for DICOM-to-BIDS conversion and validation; fMRIPrep itself runs inside the pinned Docker image.

Core rule

Do not start fMRIPrep until the BIDS Validator reports zero errors. A successful conversion is not the same as a valid BIDS dataset.

Document Conventions

Notation	Meaning
<code>/your_folder/fmri_project</code>	The project root. Replace only <code>/your_folder/</code> with an approved storage location.
<code>GAB01</code>	Example participant label. Pass labels without the sub- prefix to command-line tools.
<code>\$VARIABLE</code>	A shell variable defined earlier in the command or script.
Commands	Run in a Bash-compatible shell on the Linux server.
Pinned image	nipreps/fmriprep:25.2.5. Do not silently replace it with latest.

Workflow at a Glance

1. Confirm server access, Docker permission, disk space, and the shared fMRIPrep image.
2. Create the project directory tree and a dedicated Conda environment.
3. Place one participant's DICOM files in sourcedata and run `dcm2bids_helper`.
4. Inspect the helper JSON sidecars and build a study-specific `dcm2bids` configuration.
5. Convert each participant to BIDS and add dataset-level files.
6. Run the BIDS Validator and resolve every error.
7. Verify fieldmap links and metadata before preprocessing.
8. Run fMRIPrep in `tmux`, save logs, and preserve the work directory until success.
9. Inspect the HTML report and perform study-specific quality control.
10. Record software versions, image digest, commands, and decisions.

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1. Assumptions and Responsibilities

This workflow separates administrator responsibilities from student responsibilities.

Role	Responsibilities
Server administrator / lab manager	Install and maintain Docker Engine; pull the approved fMRIPrep image; manage Docker-group access; maintain secure storage; provide a readable FreeSurfer license; monitor server capacity.
Student user	Create the project folders and Conda environment; convert and validate data; run fMRIPrep with approved resource limits; inspect reports; document all commands and versions.
Study lead	Approve BIDS naming, fieldmap strategy, preprocessing options, QC criteria, and data-release/de-identification procedures.

Security warning

Membership in the docker group generally grants root-level privileges on the server. Students must not be added without administrator approval, and they must not mount or modify directories outside the approved project area.

1.1 Confirm Login, Shell, Time, and Storage

```
whoami
hostname
date
pwd

df -h /your_folder
free -h
nproc
```

Record the server name and date in the preprocessing log. Ensure that the project filesystem has enough free space. fMRIPrep work directories can be much larger than the final derivatives, particularly for high-resolution anatomical data and many BOLD runs.

1.2 Confirm Conda

```
conda --version
which conda
```

If conda is not found

Do not install a second system-wide Conda distribution without permission. Ask the server administrator for the correct initialization command or module.

2. Create the Project Directory Tree

Use a consistent layout. Keep DICOM source data, the BIDS dataset, final derivatives, disposable work files, code, logs, helper output, and TemplateFlow cache in separate directories.

```
export PROJECT_ROOT=/your_folder/fmri_project

mkdir -p \
  "$PROJECT_ROOT/sourcedata/dicom" \
  "$PROJECT_ROOT/rawdata" \
  "$PROJECT_ROOT/derivatives/fmrip-25.2.5" \
  "$PROJECT_ROOT/work/fmrip-25.2.5" \
  "$PROJECT_ROOT/code" \
  "$PROJECT_ROOT/logs" \
  "$PROJECT_ROOT/helper" \
  "$PROJECT_ROOT/templateflow" \
  "$PROJECT_ROOT/licenses"

find "$PROJECT_ROOT" -maxdepth 2 -type d | sort
```

Directory	Purpose	Keep after completion?
sourcedata/dicom	Original scanner exports. Treat as sensitive source data.	Yes, according to lab policy
rawdata	Valid BIDS dataset used as fMRIPrep input.	Yes
derivatives/fmrip-25.2.5	Final fMRIPrep outputs and reports.	Yes
work/fmrip-25.2.5	Large intermediate files used for resume and debugging.	Until successful QC and backup
code	Configuration files, scripts, environment file, version records.	Yes
logs	Conversion, validation, and fMRIPrep terminal logs.	Yes
helper	Temporary NIfTI/JSON sidecars from dcm2bids_helper.	Optional after config is finalized
templateflow	Shared or project-local template cache.	Yes; reusable

Important directory rule

Do not run dcm2bids_helper inside the BIDS rawdata directory. Its tmp_dcm2bids folder is not a BIDS data type and can trigger a NOT_INCLUDED validation error.

3. Create the Conda Environment

The environment below contains dcm2bids, dcm2niix, jq, and the command-line BIDS Validator. It does not install fMRIPrep; fMRIPrep is provided by Docker.

3.1 Create environment.yml

```
cat > /your_folder/fmri_project/code/environment.yml <<'EOF'
name: fmri-bids
channels:
  - conda-forge
dependencies:
  - python=3.11
  - dcm2bids=3.2.0
  - dcm2niix
  - jq
  - pip
  - pip:
    - bids-validator-deno
EOF
```

3.2 Create and Activate the Environment

```
cd /your_folder/fmri_project/code
conda env create --file environment.yml
conda activate fmri-bids
```

When the environment already exists and environment.yml has changed, update it reproducibly:

```
conda env update --file environment.yml --prune
conda activate fmri-bids
```

3.3 Verify Every Tool

```
python --version
dcm2bids --version
dcm2niix --version
jq --version
bids-validator-deno --help | head
```

Expected result

Each command should print a version or help text without a command-not-found error. Save exact versions later in code/software_versions.txt.

4. Verify Docker and the Shared fMRIPrep Image

Docker images are managed by the shared Docker daemon, not downloaded into the student's current directory. If the administrator has already pulled nipreps/fmriprep:25.2.5, all authorized users of the same daemon can use it.

4.1 Check Docker Access

```
id -nG
docker info >/dev/null && echo "Docker access: OK"
```

Permission denied at /var/run/docker.sock

Your current login session does not have Docker access. Ask the administrator to confirm docker-group membership, then log out completely and log back in. Do not work around the problem by repeatedly using sudo unless that is the server's approved policy.

4.2 Check the Pinned Image

```
docker image inspect nipreps/fmriprep:25.2.5 >/dev/null \  
&& echo "fMRIPrep image found"  
  
docker run --rm nipreps/fmriprep:25.2.5 --version
```

If the image is missing, the administrator can pull it once for the whole server:

```
docker pull nipreps/fmriprep:25.2.5
```

Reproducibility

Always use the explicit tag nipreps/fmriprep:25.2.5. Avoid latest because it can change without any change to your script.

5. Prepare DICOM Source Data

5.1 Choose a Participant Label

Use an alphanumeric participant label that does not contain spaces or personal identifiers. The example label is GAB01. Command-line tools receive GAB01, not sub-GAB01. BIDS filenames will include sub-GAB01 automatically.

```
export SUBJECT=GAB01
export PROJECT_ROOT=/your_folder/fmri_project
export DICOM_DIR="$PROJECT_ROOT/sourcedata/dicom/sub-$SUBJECT"
```

5.2 Copy or Move the Scanner Export

```
mkdir -p "$DICOM_DIR"

# Example only: copy a participant export into the project.
# Replace /path/from_scanner with the approved secure source path.
rsync -av --progress /path/from_scanner/ "$DICOM_DIR/"

find "$DICOM_DIR" -type f | head
find "$DICOM_DIR" -type f | wc -l
```

Protected health information

Raw DICOM headers may contain identifying information. Store and process DICOM data only on approved secure systems. BIDS validation does not certify de-identification. Defacing and metadata review are separate responsibilities before data sharing.

5.3 Preserve the Original Source

- Do not edit DICOM files in place.
- Keep one immutable copy under sourcedata according to lab retention policy.
- Document the scanner, sequence version, acquisition date policy, and any transfer or anonymization tool.
- Do not include participant names, dates of birth, medical record numbers, or other direct identifiers in directory names.

6. Generate Helper Sidcars

Use `dcm2bids_helper` on one representative participant. The helper runs `dcm2niix` and produces NIfTI files and JSON sidecars that reveal the metadata needed to construct the configuration file.

```
conda activate fmri-bids

export PROJECT_ROOT=/your_folder/fmri_project
export SUBJECT=GAB01
export DICOM_DIR="$PROJECT_ROOT/sourcedata/dicom/sub-$SUBJECT"
export HELPER_DIR="$PROJECT_ROOT/helper/sub-$SUBJECT"

rm -rf "$HELPER_DIR"
mkdir -p "$HELPER_DIR"

dcm2bids_helper \
  -d "$DICOM_DIR" \
  -o "$HELPER_DIR" \
  2>&1 | tee "$PROJECT_ROOT/logs/dcm2bids_helper_sub-${SUBJECT}.log"
```

6.1 Locate the JSON Sidcars

```
find "$HELPER_DIR/tmp_dcm2bids/helper" \
  -maxdepth 1 -type f -name '*.json' | sort
```

6.2 Summarize Important Metadata

```
for f in "$HELPER_DIR"/tmp_dcm2bids/helper/*.json; do
  echo "==== $(basename "$f") ====="
  jq '{
    SeriesNumber,
    SeriesDescription,
    ProtocolName,
    ImageType,
    RepetitionTime,
    EchoTime,
    PhaseEncodingDirection,
    TotalReadoutTime
  }' "$f"
done | tee "$PROJECT_ROOT/logs/helper_metadata_sub-${SUBJECT}.txt"
```

What to identify

Identify the exact SeriesDescription or ProtocolName for T1w, each BOLD run, each SBRef, and every AP/PA distortion-correction acquisition. Confirm that PhaseEncodingDirection and TotalReadoutTime are present in the functional and fieldmap JSON files.

7. Build the dcm2bids Configuration

The configuration maps scanner metadata to BIDS filenames. Exact scanner descriptions vary across protocols; therefore, the example below must be checked against the helper JSON files. Criteria use shell-style matching, and all criteria in a description must match.

Do not copy blindly

A configuration that works for one protocol may silently misclassify a different protocol. Compare every SeriesDescription with the helper output before converting participants.

7.1 Example Study Configuration

```
cat > /your_folder/fmri_project/code/dcm2bids_config.json <<'EOF'
{
  "descriptions": [
    {
      "datatype": "anat",
      "suffix": "T1w",
      "criteria": {
        "SeriesDescription": "T1_mprage_sag_0.7iso"
      }
    },
    {
      "datatype": "func",
      "suffix": "sbref",
      "custom_entities": "task-rest",
      "criteria": {
        "SeriesDescription": "A-P_REST_Muitiband8_epi_cmrr_SBRef"
      },
      "sidecar_changes": {
        "TaskName": "rest",
        "B0FieldSource": "pepolar"
      }
    },
    {
      "datatype": "func",
      "suffix": "bold",
      "custom_entities": "task-rest",
      "criteria": {
        "SeriesDescription": "A-P_REST_Muitiband8_epi_cmrr"
      },
      "sidecar_changes": {
        "TaskName": "rest",
        "B0FieldSource": "pepolar"
      }
    },
    {
      "datatype": "func",
      "suffix": "sbref",
      "custom_entities": "task-motor_run-01",
      "criteria": {
        "SeriesDescription": "RUN1_Muitiband8_epi_cmrr_SBRef"
      },
      "sidecar_changes": {
        "TaskName": "motor",
        "B0FieldSource": "pepolar"
      }
    },
    {
      "datatype": "func",
      "suffix": "bold",
      "custom_entities": "task-motor_run-01",
      "criteria": {
        "SeriesDescription": "RUN1_Muitiband8_epi_cmrr"
      },
      "sidecar_changes": {
        "TaskName": "motor",
        "B0FieldSource": "pepolar"
      }
    },
    {
      "datatype": "fmap",
      "suffix": "epi",
      "custom_entities": "dir-AP",
      "criteria": {
```

```

    "SeriesDescription":
      "distortion_corr_64ch_pa_polarity_invert_to_ap"
  },
  "sidecar_changes": {
    "B0FieldIdentifier": "pepolar"
  }
},
{
  "datatype": "fmap",
  "suffix": "epi",
  "custom_entities": "dir-PA",
  "criteria": {
    "SeriesDescription": "distortion_corr_64ch_pa"
  },
  "sidecar_changes": {
    "B0FieldIdentifier": "pepolar"
  }
}
]
}
EOF

```

7.2 Validate the JSON Syntax

```

jq empty /your_folder/fmri_project/code/dcm2bids_config.json \
&& echo "Configuration JSON syntax: OK"

```

7.3 Understand the Fieldmap Link

Both opposite-phase EPI fieldmaps carry the same `B0FieldIdentifier`, `pepolar`. The functional images carry `B0FieldSource` with the same value. This declares that the paired fieldmap estimation may be used to correct those functional acquisitions. The `dir-AP` and `dir-PA` labels are human-readable acquisition labels; the JSON `PhaseEncodingDirection` value is authoritative.

Files	Metadata	Role
<code>fmap/sub-*_dir-AP_epi.*</code>	<code>B0FieldIdentifier: pepolar</code>	One member of the opposite-PE fieldmap set
<code>fmap/sub-*_dir-PA_epi.*</code>	<code>B0FieldIdentifier: pepolar</code>	The other member of the opposite-PE fieldmap set
<code>func/sub-*_task-*_bold.*</code>	<code>B0FieldSource: pepolar</code>	Requests correction using that field estimation
<code>func/sub-*_task-*_sbref.*</code>	<code>B0FieldSource: pepolar</code>	Links the SBRef to the same field estimation

8. Convert DICOM to BIDS

8.1 Define Paths and Run Conversion

```

conda activate fmri-bids

export PROJECT_ROOT=/your_folder/fmri_project
export SUBJECT=GAB01
export DICOM_DIR="$PROJECT_ROOT/sourcedata/dicom/sub-$SUBJECT"
export BIDS_DIR="$PROJECT_ROOT/rawdata"
export CONFIG="$PROJECT_ROOT/code/dcm2bids_config.json"

mkdir -p "$BIDS_DIR"

dcm2bids \
  -d "$DICOM_DIR" \
  -p "$SUBJECT" \
  -c "$CONFIG" \
  -o "$BIDS_DIR" \
  2>&1 | tee "$PROJECT_ROOT/logs/dcm2bids_sub-${SUBJECT}.log"

```

8.2 Inspect the Result

```
find "$BIDS_DIR/sub-$_SUBJECT" -maxdepth 3 -type f | sort
```

A typical result should contain anat, func, and fmap directories:

```
sub-GAB01/
  anat/
    sub-GAB01_T1w.json
    sub-GAB01_T1w.nii.gz
  fmap/
    sub-GAB01_dir-AP_epi.json
    sub-GAB01_dir-AP_epi.nii.gz
    sub-GAB01_dir-PA_epi.json
    sub-GAB01_dir-PA_epi.nii.gz
  func/
    sub-GAB01_task-rest_bold.json
    sub-GAB01_task-rest_bold.nii.gz
    sub-GAB01_task-rest_sbref.json
    sub-GAB01_task-rest_sbref.nii.gz
    sub-GAB01_task-motor_run-01_bold.json
    sub-GAB01_task-motor_run-01_bold.nii.gz
    sub-GAB01_task-motor_run-01_sbref.json
    sub-GAB01_task-motor_run-01_sbref.nii.gz
```

8.3 Review dcm2bids Mismatches

dcm2bids places unmatched or multiply matched files in rawdata/tmp_dcm2bids. Review the logs and temporary sidecars. Fix the configuration rather than ignoring unexpected files.

```
find "$BIDS_DIR/tmp_dcm2bids" -maxdepth 3 -type f 2>/dev/null | sort

grep -E "No Pairing|more than one|not paired|WARNING|ERROR" \
  "$PROJECT_ROOT/logs/dcm2bids_sub-$_SUBJECT.log" || true
```

After successful review

Remove rawdata/tmp_dcm2bids before final BIDS validation. It is conversion workspace, not raw BIDS data.

```
rm -rf "$BIDS_DIR/tmp_dcm2bids"
```

8.4 Safe Re-conversion

Never delete a multi-participant dataset

When re-converting one participant, remove only that participant's BIDS directory and conversion workspace. Confirm the SUBJECT variable before running rm -rf.

```
printf 'SUBJECT=%s\nBIDS_DIR=%s\n' "$SUBJECT" "$BIDS_DIR"

# Use only after visually checking the two lines above.
rm -rf "$BIDS_DIR/sub-$_SUBJECT" "$BIDS_DIR/tmp_dcm2bids"

# Then rerun the dcm2bids command.
```

9. Add BIDS Dataset-Level Files

9.1 dataset_description.json

```
cat > /your_folder/fmri_project/rawdata/dataset_description.json <<'EOF'
{
  "Name": "Your fMRI Study",
  "BIDSVersion": "1.10.0",
  "DatasetType": "raw",
  "Authors": [
    "Your Laboratory"
  ]
}
EOF

jq . /your_folder/fmri_project/rawdata/dataset_description.json
```

9.2 README

```
cat > /your_folder/fmri_project/rawdata/README <<'EOF'
Your fMRI Study
=====

This directory contains the raw BIDS dataset.
DICOM source data are stored separately under sourcedata/.
Preprocessed derivatives are stored under derivatives/.
EOF
```

9.3 participants.tsv

```
cat > /your_folder/fmri_project/rawdata/participants.tsv <<'EOF'
participant_id
sub-GAB01
EOF
```

Add one row per participant. Do not add sensitive demographic fields unless they are required, approved, and properly governed.

9.4 Task Events

fMRIPrep does not use event timing for preprocessing, but task-based statistical analysis normally requires a task events TSV file. Create it from the experimental log, not from memory. Resting-state runs do not require events.

```
cat > /your_folder/fmri_project/rawdata/sub-GAB01/func/\
sub-GAB01_task-motor_run-01_events.tsv <<'EOF'
onset   duration   trial_type
0.0     2.0           movement
4.0     2.0           movement
EOF
```

Timing convention

Onsets are measured in seconds relative to the beginning of the first stored BOLD volume. Confirm trigger handling, discarded volumes, and synchronization with the acquisition protocol.

10. Validate and Inspect the BIDS Dataset

10.1 Run the BIDS Validator

```
conda activate fmri-bids
```

```
export PROJECT_ROOT=/your_folder/fmri_project
export BIDS_DIR="$PROJECT_ROOT/rawdata"

bids-validator-deno "$BIDS_DIR" \
  2>&1 | tee "$PROJECT_ROOT/logs/bids_validation.txt"
```

Pass criterion

Errors must equal zero. Warnings must be read and documented; they are not automatically harmless.

10.2 Common NOT_INCLUDED Error

```
# Typical cause:
/your_folder/fmri_project/rawdata/tmp_dcm2bids/

# Fix after reviewing conversion mismatches:
rm -rf /your_folder/fmri_project/rawdata/tmp_dcm2bids
bids-validator-deno /your_folder/fmri_project/rawdata
```

10.3 Verify Functional and Fieldmap Metadata

```
export SUBJECT=GAB01
export BIDS_DIR=/your_folder/fmri_project/rawdata

grep -R -E \
  '"TaskName"|"B0FieldSource"|"B0FieldIdentifier"|"PhaseEncodingDirection"|"TotalReadoutTime"' \
  "$BIDS_DIR/sub-$_SUBJECT"
```

10.4 Programmatic Metadata Checks

```
for f in "$BIDS_DIR/sub-$_SUBJECT"/func/*.json \
  "$BIDS_DIR/sub-$_SUBJECT"/fmap/*.json; do
  echo "==== $(basename "$f") ====="
  jq '{
    TaskName,
    PhaseEncodingDirection,
    TotalReadoutTime,
    B0FieldSource,
    B0FieldIdentifier
  }' "$f"
done
```

Phase-encoding check

The two fieldmap EPI files must have opposite PhaseEncodingDirection values, such as j and j-. The dir-AP and dir-PA filename labels alone do not prove the polarity. Check the JSON values and the acquisition protocol.

10.5 Basic NIfTI Sanity Checks

Open T1w, BOLD, SBRef, and fieldmap images in a neuroimaging viewer. Confirm orientation, brain coverage, expected volume count, and the absence of obvious transfer or reconstruction failures. Automated validation cannot detect all image-content problems.

- T1w: full head/brain coverage and no severe motion artifact.
- BOLD: expected number of time points, coverage, and orientation.
- SBRef: geometrically consistent with its corresponding BOLD run.
- AP/PA fieldmaps: same geometry and opposite distortion polarity.
- No accidental duplicate or localizer series in the BIDS dataset.

11. Prepare the FreeSurfer License and Caches

11.1 Choose the License Path

On the example shared server, a readable license may already exist at `/usr/local/freesurfer/7.4.1/license.txt`. A project-local alternative is `/your_folder/fmri_project/licenses/license.txt`. Use only a license approved by the lab or server administrator.

```
export FS_LICENSE=/usr/local/freesurfer/7.4.1/license.txt

test -r "$FS_LICENSE" \
  && echo "FreeSurfer license is readable" \
  || echo "FreeSurfer license is missing or unreadable"

ls -l "$FS_LICENSE"
```

Do not print or publish license contents

The workflow only needs read access to the file. Avoid pasting the license key into public repositories, screenshots, or shared documents.

11.2 Prepare Output, Work, and TemplateFlow Directories

```
export PROJECT_ROOT=/your_folder/fmri_project

mkdir -p \
  "$PROJECT_ROOT/derivatives/fmriprep-25.2.5" \
  "$PROJECT_ROOT/work/fmriprep-25.2.5" \
  "$PROJECT_ROOT/templateflow" \
  "$PROJECT_ROOT/logs"

test -w "$PROJECT_ROOT/derivatives/fmriprep-25.2.5" \
  && echo "Output directory writable"
test -w "$PROJECT_ROOT/work/fmriprep-25.2.5" \
  && echo "Work directory writable"
```

TemplateFlow templates may be downloaded on the first run. Mounting a persistent templateflow directory prevents repeated downloads and makes template versions easier to audit.

12. Run fMRIPrep Safely in tmux

12.1 Why tmux Is Used

fMRIPrep may run for many hours. tmux keeps the server-side session alive if VS Code Remote-SSH disconnects, the laptop sleeps, or the local network changes.

```
tmux new -s fmriprep_GAB01
```

- **Detach without stopping:** press `Ctrl+b`, release the keys, then press `d`.
- **List sessions:** `tmux ls`
- **Reattach:** `tmux attach -t fmriprep_GAB01`
- **Leave a finished session:** type `exit` after the command completes.

12.2 Full fMRIPrep Command

```
export PROJECT_ROOT=/your_folder/fmri_project
export SUBJECT=GAB01
export BIDS_DIR="$PROJECT_ROOT/rawdata"
export OUT_DIR="$PROJECT_ROOT/derivatives/fmriprep-25.2.5"
export WORK_DIR="$PROJECT_ROOT/work/fmriprep-25.2.5"
export TEMPLATEFLOW_DIR="$PROJECT_ROOT/templateflow"
export FS_LICENSE=/usr/local/freesurfer/7.4.1/license.txt
```

```

export IMAGE=nipreps/fmriprep:25.2.5

mkdir -p "$OUT_DIR" "$WORK_DIR" "$TEMPLATEFLOW_DIR" \
"$PROJECT_ROOT/logs"

set -o pipefail

docker run --rm \
  --user "$(id -u):$(id -g)" \
  -e HOME=/tmp \
  -e TEMPLATEFLOW_HOME=/templateflow \
  -v "$BIDS_DIR":/data:ro \
  -v "$OUT_DIR":/out \
  -v "$WORK_DIR":/work \
  -v "$TEMPLATEFLOW_DIR":/templateflow \
  -v "$FSL_LICENSE":/opt/freesurfer/license.txt:ro \
  "$IMAGE" \
  /data /out participant \
  --participant-label "$SUBJECT" \
  --fs-license-file /opt/freesurfer/license.txt \
  --output-spaces MNI152NLin2009cAsym:res-2 T1w \
  --nprocs 8 \
  --omp-nthreads 4 \
  --mem-mb 24000 \
  -w /work \
  --notrack \
  2>&1 | tee "$PROJECT_ROOT/logs/fmriprep_sub-${SUBJECT}.log"

```

12.3 What the Main Options Mean

Option	Meaning
--user UID:GID	Writes outputs as the current student rather than as root.
/data:ro	BIDS input is mounted read-only so fMRIPrep cannot alter raw data.
/out	Final derivatives and HTML report.
/work	Intermediate workflow files used for resume and debugging.
TEMPLATEFLOW_HOME	Persistent template cache mounted from the project directory.
--participant-label GAB01	Processes only sub-GAB01; do not include the sub- prefix.
--output-spaces	Writes volumetric results in individual T1w and 2 mm MNI152NLin2009cAsym spaces.
--nprocs 8	Maximum threads across processes.
--omp-nthreads 4	Maximum OpenMP threads per process.
--mem-mb 24000	Upper memory bound of 24,000 MB.
--notrack	Opts out of run-tracking telemetry.

Coordinate resource use

The example limits are not a license to occupy the whole server. Check current load and follow lab scheduling rules. Multiple students must not independently oversubscribe CPU, memory, or disk I/O.

13. Monitor, Resume, and Complete the Run

13.1 Monitor from Another Terminal

```
docker ps
tail -f /your_folder/fmri_project/logs/fmriprep_sub-GAB01.log
du -sh /your_folder/fmri_project/work/fmriprep-25.2.5
df -h /your_folder
```

Exit tail -f with Ctrl+c; this does not stop fMRIPrep.

13.2 Confirm Whether tmux Is Attached

```
tmux ls
```

The word attached means that a client is currently viewing the session. detached means the session continues in the background without a connected viewer.

13.3 Resume After a Crash or Disconnect

If the container stopped but the work directory is intact, run the same command again with the same BIDS, output, work, image, and participant settings. fMRIPrep can reuse completed workflow nodes from the existing work directory.

Do not clean the work directory too early

Keep the work directory until the run completes, the HTML report passes QC, outputs are backed up, and no further resume or debugging is expected.

13.4 Detect Completion

```
export PROJECT_ROOT=/your_folder/fmri_project
export SUBJECT=GAB01
export OUT_DIR="$PROJECT_ROOT/derivatives/fmriprep-25.2.5"

ls -lh "$OUT_DIR/sub-${SUBJECT}.html"
find "$OUT_DIR/sub-${SUBJECT}" -type f | wc -l

tail -n 40 "$PROJECT_ROOT/logs/fmriprep_sub-${SUBJECT}.log"
```

Completion is not quality control

A zero exit code and an HTML file mean the workflow finished, not that every registration, mask, or distortion correction is scientifically acceptable.

14. Inspect fMRIPrep Outputs and Quality-Control Reports

14.1 Key Output Locations

Output	Typical location or pattern	Use
Subject report	derivatives/fmriprep-25.2.5/sub-GAB01.html	Primary visual QC report
Preprocessed BOLD	sub-GAB01/func/*_desc-preproc_bold.nii.gz	First-level analysis input
Brain masks	sub-GAB01/func/*_desc-brain_mask.nii.gz	Masking and QC
Confounds	sub-GAB01/func/*_desc-confounds_timeseries.tsv	Motion/physiology/nuisance modeling
Transforms	sub-GAB01/anat/*_xfm.*	Movement between anatomical/template spaces
Anatomical derivatives	sub-GAB01/anat/	Preprocessed T1w, masks, segmentations

Output	Typical location or pattern	Use
FreeSurfer outputs	freesurfer/sub-GAB01/	Surface reconstruction and parcellations
Citation boilerplate	logs/CITATION.*	Methods and software citation text

14.2 Open the HTML Report

Download or open the subject HTML report through an approved method. Review every section, not only the summary banner. Preserve the report with the derivatives.

14.3 Minimum Visual QC Checklist

QC item	Question
Anatomical brain extraction	Brain tissue retained; skull and non-brain tissue adequately excluded.
Anatomical segmentation	Gray matter, white matter, and CSF align with the T1w anatomy.
T1w-to-template registration	Cortical and subcortical structures align with the template overlay.
BOLD reference and mask	Expected coverage; no severe cropping or mask failure.
BOLD-to-T1w registration	Functional boundaries align with anatomical structures.
Susceptibility distortion correction	Corrected EPI is improved and not over-warped; AP/PA fieldmap usage is reported.
Motion and confounds	Review framewise displacement, DVARS, outlier volumes, and carpet plots.
Warnings	Read every warning and decide whether it affects inclusion, reprocessing, or analysis.

14.4 Confounds Are Not a Universal Denoising Recipe

fMRIPrep estimates many confound regressors but does not choose a single optimal nuisance-regression strategy for every study. The analysis plan must predefine which motion, CompCor, cosine, outlier, and other regressors will be used. Do not select confounds after seeing the desired statistical result.

14.5 What fMRIPrep Does Not Finish

- Spatial smoothing for a specific statistical analysis.
- Task GLM design and contrast estimation.
- Resting-state denoising strategy and connectivity estimation.
- Group-level statistics.
- Participant exclusion decisions.
- Scientific interpretation of warnings or image artifacts.

15. Process Multiple Participants

15.1 Convert Participants One at a Time

Use the same validated configuration only when the acquisition protocol is consistent. Spot-check helper metadata for early participants and whenever the scanner protocol changes.

```
for SUBJECT in GAB01 GAB02 GAB03; do
  /your_folder/fmri_project/code/convert_subject.sh "$SUBJECT"
done
```

15.2 Sequential fMRIPrep Batch

Sequential processing is safer for a shared server because it limits resource contention.

```
for SUBJECT in GAB01 GAB02 GAB03; do
  /your_folder/fmri_project/code/run_fmriprep.sh "$SUBJECT"
done
```

15.3 Parallel Processing

Administrator approval required

Before running participants in parallel, calculate the combined `--nprocs` and `--mem-mb` values, check available RAM and disk I/O, and coordinate with other users. Two jobs configured for 24 GB each may require substantially more than 48 GB of safe system capacity when overhead and other users are included.

15.4 One tmux Session per Participant

```
tmux new -s fmriprep_GAB01
# Run: /your_folder/fmri_project/code/run_fmriprep.sh GAB01

# In a separate terminal, only when approved:
tmux new -s fmriprep_GAB02
# Run: /your_folder/fmri_project/code/run_fmriprep.sh GAB02
```

16. Reproducibility and Provenance

16.1 Record Software Versions and Image Identity

```
export PROJECT_ROOT=/your_folder/fmri_project

{
  echo "Date: $(date -Is)"
  echo "Host: $(hostname)"
  echo "User: $(whoami)"
  echo
  conda --version
  conda run -n fmri-bids python --version
  conda run -n fmri-bids dcm2bids --version
  conda run -n fmri-bids dcm2niix --version
  conda run -n fmri-bids bids-validator-deno --version || true
  docker --version
  docker run --rm nipreps/fmriprep:25.2.5 --version
  docker image inspect \
    --format 'Image ID: {{.Id}}' \
    nipreps/fmriprep:25.2.5
  docker image inspect \
    --format 'Repo digests: {{json .RepoDigests}}' \
    nipreps/fmriprep:25.2.5
} | tee "$PROJECT_ROOT/code/software_versions.txt"
```

16.2 Export the Conda Environment

```
conda env export -n fmri-bids --from-history \
> /your_folder/fmri_project/code/environment_from_history.yml

conda list -n fmri-bids --explicit \
> /your_folder/fmri_project/code/conda_explicit_spec.txt
```

16.3 Preserve Exact Commands

- Keep dcm2bids_config.json under version control.
- Keep environment.yml and the explicit package specification.
- Keep conversion, validator, and fMRIPrep logs for every participant.
- Record any rerun, manual metadata correction, exclusion, or non-default fMRIPrep option.
- Use the generated fMRIPrep CITATION files when writing the Methods section.

17. Storage, Privacy, and Permissions

17.1 Recommended Permission Model

Use a lab Unix group and shared group ownership rather than world-writable permissions. Exact commands depend on server policy and should be run or approved by the administrator.

```
# Administrator example only - do not run without approval.
chgrp -R your_lab_group /your_folder/fmri_project
chmod -R g+rwX /your_folder/fmri_project
find /your_folder/fmri_project -type d -exec chmod g+s {} +
```

Avoid chmod 777

World-writable project directories are not an acceptable fix for permission problems. Diagnose ownership, group membership, umask, and Docker user mapping instead.

17.2 Backup Priorities

11. BIDS rawdata and dataset-level metadata.
12. Code, configuration files, environment specifications, and version records.
13. fMRIPrep derivatives and HTML reports.
14. Logs and QC decisions.
15. DICOM source data according to institutional policy.
16. Work directory only while resume/debugging is still needed.

17.3 Cleanup Only After Approval

```
# Review size first.
du -sh /your_folder/fmri_project/work/fmriprep-25.2.5

# Remove only after successful QC, backup, and study-lead approval.
# rm -rf /your_folder/fmri_project/work/fmriprep-25.2.5
```

18. Troubleshooting

Problem	Action
Docker permission denied	Run <code>id -nG</code> . Ask the administrator to confirm docker-group membership. Log out completely and reconnect. A pre-existing tmux shell may retain old group membership.
fMRIPrep image not found	Run <code>docker image inspect nipreps/fmriprep:25.2.5</code> . Ask the administrator to pull the pinned image.
BIDS NOT_INCLUDED: tmp_dcm2bids	Review conversion mismatches, then remove <code>rawdata/tmp_dcm2bids</code> and validate again.
No DICOM sidecars matched	Compare criteria with helper JSON. Check exact SeriesDescription, capitalization, and wildcard use.
One acquisition matched multiple descriptions	Make criteria more specific, often by adding ImageType, ProtocolName, or another stable metadata key.
Missing TaskName	Add TaskName through <code>sidecar_changes</code> for BOLD and SBRef descriptions, then reconvert.
Fieldmap not used	Confirm opposite PhaseEncodingDirection values, TotalReadoutTime, shared B0FieldIdentifier, and matching B0FieldSource.
FreeSurfer license error	Confirm test -r on the host path and the read-only Docker mount to <code>/opt/freesurfer/license.txt</code> .
Root-owned output files	Use <code>--user \$(id -u):\$(id -g)</code> . Repair ownership only with administrator approval.
TemplateFlow download failure	Check network access, free space, and write permission on the mounted templateflow directory. Retry with the same cache.
Killed / exit 137	Likely out-of-memory. Inspect system logs if available, reduce concurrency, and coordinate memory limits with the administrator.
No space left on device	Check <code>df -h</code> and <code>du -sh</code> . Work files are often the largest component. Do not delete an active work directory.
tmux session already attached	Use <code>tmux attach -d -t SESSION</code> to detach the other client and attach here, but first confirm another student is not actively using it.
Run stopped after SSH disconnect	The command was probably not inside tmux. Start a tmux session and rerun with the same work directory.
Validation passes but images are wrong	BIDS validation checks structure and metadata consistency, not full image quality. Perform visual inspection and protocol-level checks.

18.1 Rapid Diagnostic Bundle

```
export PROJECT_ROOT=/your_folder/fmri_project
export SUBJECT=GAB01

{
  date -Is
  hostname
  whoami
  id
  df -h /your_folder
  free -h
  docker --version
  docker ps -a
  docker image inspect nipreps/fmriprep:25.2.5 >/dev/null \
    && echo "Image present"
  tail -n 100 \
    "$PROJECT_ROOT/logs/fmriprep_sub-${SUBJECT}.log"
} > "$PROJECT_ROOT/logs/diagnostic_sub-${SUBJECT}.txt" 2>&1
```

Before sharing diagnostics

Review logs for participant identifiers, protected information, license content, private paths, and other sensitive details. Share only through approved channels.

Appendix A. Complete environment.yml

```
name: fmri-bids
channels:
  - conda-forge
dependencies:
  - python=3.11
  - dcm2bids=3.2.0
  - dcm2niix
  - jq
  - pip
  - pip:
    - bids-validator-deno
```

Appendix B. Complete dcm2bids_config.json

```
{
  "descriptions": [
    {
      "datatype": "anat",
      "suffix": "T1w",
      "criteria": {
        "SeriesDescription": "T1_mprage_sag_0.7iso"
      }
    },
    {
      "datatype": "func",
      "suffix": "sbref",
      "custom_entities": "task-rest",
      "criteria": {
        "SeriesDescription": "A-P_REST_Muitiband8_epi_cmrr_SBRef"
      },
      "sidecar_changes": {
        "TaskName": "rest",
        "B0FieldSource": "pepolar"
      }
    },
    {
      "datatype": "func",
      "suffix": "bold",
      "custom_entities": "task-rest",
      "criteria": {
        "SeriesDescription": "A-P_REST_Muitiband8_epi_cmrr"
      },
      "sidecar_changes": {
        "TaskName": "rest",
        "B0FieldSource": "pepolar"
      }
    },
    {
      "datatype": "func",
      "suffix": "sbref",
      "custom_entities": "task-motor_run-01",
      "criteria": {
        "SeriesDescription": "RUN1_Muitiband8_epi_cmrr_SBRef"
      },
      "sidecar_changes": {
        "TaskName": "motor",
        "B0FieldSource": "pepolar"
      }
    },
    {
      "datatype": "func",
      "suffix": "bold",
      "custom_entities": "task-motor_run-01",
      "criteria": {
        "SeriesDescription": "RUN1_Muitiband8_epi_cmrr"
      },
      "sidecar_changes": {
        "TaskName": "motor",
        "B0FieldSource": "pepolar"
      }
    }
  ]
}
```

```

    }
  },
  {
    "datatype": "fmap",
    "suffix": "epi",
    "custom_entities": "dir-AP",
    "criteria": {
      "SeriesDescription":
        "distortion_corr_64ch_pa_polarity_invert_to_ap"
    },
    "sidecar_changes": {
      "B0FieldIdentifier": "pepolar"
    }
  },
  {
    "datatype": "fmap",
    "suffix": "epi",
    "custom_entities": "dir-PA",
    "criteria": {
      "SeriesDescription": "distortion_corr_64ch_pa"
    },
    "sidecar_changes": {
      "B0FieldIdentifier": "pepolar"
    }
  }
]
}

```

Appendix C. Executable Project Scripts

The scripts below are designed for Bash. Create each file under `/your_folder/fmri_project/code` and make it executable with `chmod +x`. Read and adapt all study-specific variables before use.

C.1 setup_project.sh

```
cat > /your_folder/fmri_project/code/setup_project.sh <<'EOF'
#!/usr/bin/env bash
set -euo pipefail

PROJECT_ROOT="${1:-/your_folder/fmri_project}"

mkdir -p \
  "$PROJECT_ROOT/sourcedata/dicom" \
  "$PROJECT_ROOT/rawdata" \
  "$PROJECT_ROOT/derivatives/fmriprep-25.2.5" \
  "$PROJECT_ROOT/work/fmriprep-25.2.5" \
  "$PROJECT_ROOT/code" \
  "$PROJECT_ROOT/logs" \
  "$PROJECT_ROOT/helper" \
  "$PROJECT_ROOT/templateflow" \
  "$PROJECT_ROOT/licenses"

echo "Project created at: $PROJECT_ROOT"
find "$PROJECT_ROOT" -maxdepth 2 -type d | sort
EOF
chmod +x /your_folder/fmri_project/code/setup_project.sh
```

C.2 run_helper.sh

```
cat > /your_folder/fmri_project/code/run_helper.sh <<'EOF'
#!/usr/bin/env bash
set -euo pipefail

SUBJECT="${1:?Usage: $0 SUBJECT_LABEL}"
PROJECT_ROOT="${PROJECT_ROOT:-/your_folder/fmri_project}"
DICOM_DIR="$PROJECT_ROOT/sourcedata/dicom/sub-$SUBJECT"
HELPER_DIR="$PROJECT_ROOT/helper/sub-$SUBJECT"
LOG="$PROJECT_ROOT/logs/dcm2bids_helper_sub-${SUBJECT}.log"

[[ -d "$DICOM_DIR" ]] || {
  echo "ERROR: DICOM directory not found: $DICOM_DIR" >&2
  exit 1
}

mkdir -p "$PROJECT_ROOT/logs"
rm -rf "$HELPER_DIR"
mkdir -p "$HELPER_DIR"

dcm2bids_helper -d "$DICOM_DIR" -o "$HELPER_DIR" \
  2>&1 | tee "$LOG"

echo "Helper output: $HELPER_DIR/tmp_dcm2bids/helper"
EOF
chmod +x /your_folder/fmri_project/code/run_helper.sh
```

C.3 convert_subject.sh

```
cat > /your_folder/fmri_project/code/convert_subject.sh <<'EOF'
#!/usr/bin/env bash
set -euo pipefail

SUBJECT="${1:?Usage: $0 SUBJECT_LABEL}"
PROJECT_ROOT="${PROJECT_ROOT:-/your_folder/fmri_project}"
DICOM_DIR="$PROJECT_ROOT/sourcedata/dicom/sub-$SUBJECT"
BIDS_DIR="$PROJECT_ROOT/rawdata"
CONFIG="$PROJECT_ROOT/code/dcm2bids_config.json"
LOG="$PROJECT_ROOT/logs/dcm2bids_sub-${SUBJECT}.log"
```

```

[[ -d "$DICOM_DIR" ]] || {
    echo "ERROR: DICOM directory not found: $DICOM_DIR" >&2
    exit 1
}
[[ -f "$CONFIG" ]] || {
    echo "ERROR: Config not found: $CONFIG" >&2
    exit 1
}

mkdir -p "$BIDS_DIR" "$PROJECT_ROOT/logs"
jq empty "$CONFIG"

dcm2bids \
    -d "$DICOM_DIR" \
    -p "$SUBJECT" \
    -c "$CONFIG" \
    -o "$BIDS_DIR" \
    2>&1 | tee "$LOG"

echo
if [[ -d "$BIDS_DIR/tmp_dcm2bids" ]]; then
    echo "Review before deleting: $BIDS_DIR/tmp_dcm2bids"
fi

echo "Converted participant: sub-$SUBJECT"
EOF
chmod +x /your_folder/fmri_project/code/convert_subject.sh

```

C.4 validate_bids.sh

```

cat > /your_folder/fmri_project/code/validate_bids.sh <<'EOF'
#!/usr/bin/env bash
set -euo pipefail

PROJECT_ROOT="${PROJECT_ROOT:-/your_folder/fmri_project}"
BIDS_DIR="$PROJECT_ROOT/rawdata"
LOG="$PROJECT_ROOT/logs/bids_validation.txt"

[[ -f "$BIDS_DIR/dataset_description.json" ]] || {
    echo "ERROR: Missing $BIDS_DIR/dataset_description.json" >&2
    exit 1
}

mkdir -p "$PROJECT_ROOT/logs"
bids-validator-deno "$BIDS_DIR" 2>&1 | tee "$LOG"
EOF
chmod +x /your_folder/fmri_project/code/validate_bids.sh

```

C.5 run_fmriprep.sh

```

cat > /your_folder/fmri_project/code/run_fmriprep.sh <<'EOF'
#!/usr/bin/env bash
set -euo pipefail

SUBJECT="${1:?Usage: $0 SUBJECT_LABEL}"
PROJECT_ROOT="${PROJECT_ROOT:-/your_folder/fmri_project}"
BIDS_DIR="$PROJECT_ROOT/rawdata"
OUT_DIR="$PROJECT_ROOT/derivatives/fmriprep-25.2.5"
WORK_DIR="$PROJECT_ROOT/work/fmriprep-25.2.5"
TEMPLATEFLOW_DIR="$PROJECT_ROOT/templateflow"
FS_LICENSE="${FS_LICENSE:-/usr/local/freesurfer/7.4.1/license.txt}"
IMAGE="nipreps/fmriprep:25.2.5"
LOG="$PROJECT_ROOT/logs/fmriprep_sub-${SUBJECT}.log"

[[ -d "$BIDS_DIR/sub-$SUBJECT" ]] || {
    echo "ERROR: BIDS participant not found: sub-$SUBJECT" >&2
    exit 1
}
[[ -r "$FS_LICENSE" ]] || {
    echo "ERROR: FreeSurfer license is unreadable: $FS_LICENSE" >&2
    exit 1
}
docker image inspect "$IMAGE" >/dev/null

mkdir -p "$OUT_DIR" "$WORK_DIR" "$TEMPLATEFLOW_DIR" \
    "$PROJECT_ROOT/logs"

```

```

echo "Participant: sub-$$SUBJECT"
echo "Image:      $IMAGE"
echo "BIDS:       $BIDS_DIR"
echo "Output:     $OUT_DIR"
echo "Work:       $WORK_DIR"
echo "Log:        $LOG"

set -o pipefail

docker run --rm \
  --user "$(id -u):$(id -g)" \
  -e HOME=/tmp \
  -e TEMPLATEFLOW_HOME=/templateflow \
  -v "$BIDS_DIR":/data:ro \
  -v "$OUT_DIR":/out \
  -v "$WORK_DIR":/work \
  -v "$TEMPLATEFLOW_DIR":/templateflow \
  -v "$FSL_LICENSE":/opt/freesurfer/license.txt:ro \
  "$IMAGE" \
  /data /out participant \
  --participant-label "$SUBJECT" \
  --fs-license-file /opt/freesurfer/license.txt \
  --output-spaces MNI152NLin2009cAsym:res-2 T1w \
  --nprocs 8 \
  --omp-nthreads 4 \
  --mem-mb 24000 \
  -w /work \
  --notrack \
  2>&1 | tee "$LOG"
EOF
chmod +x /your_folder/fmri_project/code/run_fmriprep.sh

```

C.6 Typical Student Command Sequence

```

conda activate fmri-bids

export PROJECT_ROOT=/your_folder/fmri_project

# 1. One representative subject: build/check the config.
$PROJECT_ROOT/code/run_helper.sh GAB01

# 2. Convert the subject.
$PROJECT_ROOT/code/convert_subject.sh GAB01

# 3. Remove tmp_dcm2bids only after reviewing mismatches.
rm -rf $PROJECT_ROOT/rawdata/tmp_dcm2bids

# 4. Validate. Errors must be zero.
$PROJECT_ROOT/code/validate_bids.sh

# 5. Start a protected long-running session.
tmux new -s fmriprep_GAB01

# 6. Inside tmux, run preprocessing.
$PROJECT_ROOT/code/run_fmriprep.sh GAB01

```

Official Documentation Used for This Manual

1. [fMRIPrep 25.2.5 Usage Notes](https://fmriprep.org/en/25.2.5/usage.html)
https://fmriprep.org/en/25.2.5/usage.html
2. [dcm2bids 3.2.0 Documentation](https://unfmontreal.github.io/Dcm2Bids/3.2.0/)
https://unfmontreal.github.io/Dcm2Bids/3.2.0/
3. [dcm2bids Installation](https://unfmontreal.github.io/Dcm2Bids/3.2.0/get-started/install/)
https://unfmontreal.github.io/Dcm2Bids/3.2.0/get-started/install/
4. [dcm2bids Configuration File Guide](https://unfmontreal.github.io/Dcm2Bids/3.2.0/how-to/create-config-file/)
https://unfmontreal.github.io/Dcm2Bids/3.2.0/how-to/create-config-file/
5. [BIDS Validator](https://github.com/bids-standard/bids-validator)
https://github.com/bids-standard/bids-validator
6. [BIDS Specification - MRI Data](https://bids-specification.readthedocs.io/en/stable/modality-specific-files/magnetic-resonance-imaging-data.html)
https://bids-specification.readthedocs.io/en/stable/modality-specific-files/magnetic-resonance-imaging-data.html
7. [Docker Engine Linux Post-installation](https://docs.docker.com/engine/install/linux-postinstall/)
https://docs.docker.com/engine/install/linux-postinstall/
8. [FreeSurfer License](https://surfer.nmr.mgh.harvard.edu/fswiki/License)
https://surfer.nmr.mgh.harvard.edu/fswiki/License

Version note

Software interfaces and BIDS requirements evolve. This manual pins fMRIPrep 25.2.5 and dcm2bids 3.2.0. Before starting a new study or upgrading a pipeline, review the official documentation and validate a representative participant from beginning to end.

Final Preprocessing Checklist

- [] DICOM files are stored securely and participant labels contain no direct identifiers.
- [] environment.yml is saved and the fmri-bids environment activates successfully.
- [] dcm2bids_helper output was inspected for every sequence type.
- [] dcm2bids_config.json matches the actual scanner metadata.
- [] T1w, BOLD, SBRef, AP/PA EPI fieldmaps, and JSON files are present as expected.
- [] dataset_description.json and participants.tsv are present.
- [] BIDS Validator reports zero errors.
- [] Opposite fieldmap PhaseEncodingDirection values and TotalReadoutTime were checked.
- [] B0FieldIdentifier and B0FieldSource links were verified.
- [] The FreeSurfer license is readable without exposing its contents.
- [] The pinned Docker image and exact version were verified.
- [] fMRIPrep was run inside tmux with approved CPU and memory limits.
- [] The full terminal log and software version record were saved.
- [] The subject HTML report was reviewed section by section.
- [] QC decisions, reruns, exclusions, and manual changes were documented.
- [] BIDS data, derivatives, code, logs, and reports were backed up.

END OF MANUAL